

These are *dynamic factors*, because they involve time-varying positions in securities that change over time. *Dynamic* factors involve taking simultaneous *long* and *short* positions, whereas we can earn the *static* equity and bond risk premiums by taking just long positions.

Dynamic factor premiums, just like their long-only counterparts, are not a free lunch. As chapter 7 explains, while dynamic factors often beat the market over long periods of time, they can grossly underperform the market during certain periods—like the 2008–2009 financial crisis. All factor risk premiums exist in the long run because they compensate the investor for bearing losses during bad times. The factors are not appropriate for everyone because factor strategies are risky.

An important concept of dynamic factors is that they remove market exposure. Optimally constructed value-growth nets out the market portfolio and is exposed to the returns of value stocks less the returns of growth stocks. Similarly, by going long winners and short losers, momentum removes the market portfolio. In practice, factor portfolios need not be constructed with an equal number of stocks or equal dollars in opposite long-short positions. (These are called *unbalanced portfolios*.) There is no need to take short positions, but the fewer the short positions, the greater the market exposure. Put another way, the fewer short positions, the greater the correlation of the factors with the market portfolio. For example, many vehicles invest in value stocks, but they usually only take long positions. Without netting out growth stocks, the main driver of returns of these funds is the market portfolio. As the number of short positions in growth stocks increases, more market movements are removed and the more the factor reflects the *difference* between value stocks and growth stocks.

Industry often uses the terms *smart beta*, *alternative beta*, or *exotic beta* for dynamic factors. I'll stick with the term “factors” because, in asset pricing theory, beta has the strict meaning of measuring exposure to a risk factor. (These risk factors actually have a beta of one with respect to themselves.) Beta measures the magnitude of the exposure to a risk factor: we invest in factors, not betas.

Factors Define Bad Times

Each factor defines a different set of bad times. Value strategies got crushed during the financial crisis. They also got slammed during the roaring 1990s Internet bull market. In the late 1990s and early 2000s, venerable value managers, like Warren Buffet of Berkshire Hathaway, Jeremy Grantham who heads GMO Asset Management, and Julian Robertson who ran the hedge fund Tiger Management, became old-fashioned laggards. Grantham almost had to sell or close his firm. Robertson did close his fund in 2000 but went on to seed the funds of many of his associates (these hedge funds are affectionately known as “tiger cubs”).

The financial crisis was a bad time for many factors. Liquidity evaporated in commercial paper markets, mortgage-backed securities markets (especially for subprime mortgages), markets involving securitized fixed income products, and